

Deployment & Hosting Guide

Quick Deployment

Local Development

1. Open `index.html` directly in browser
2. **File Access:** Use `file://` protocol or local server
3. **Test:** Start creating pets and testing features

Creating a Local Server

Option 1: Python

```
python -m http.server 8000  
# Visit: http://localhost:8000
```

Option 2: Node.js (http-server)

```
npm install -g http-server  
http-server  
# Visit: http://localhost:8080
```

Option 3: VS Code Live Server

- Install "Live Server" extension
- Right-click `index.html`
- Select "Open with Live Server"

Web Hosting Deployment

Static Hosting Services

GitHub Pages (Free)

1. Create GitHub repository
2. Push files to `gh-pages` branch
3. Enable GitHub Pages in settings
4. Access at: <https://username.github.io/repo-name>

Netlify (Free)

1. Connect GitHub repository
2. Set build command: (leave empty)

3. Set publish directory: /
4. Deploy!

Vercel (Free)

1. Import from GitHub
2. Configure project settings
3. Click Deploy
4. Get auto-generated URL

Firebase Hosting (Free tier)

1. Install Firebase CLI
2. `firebase init hosting`
3. `firebase deploy`
4. Access at Firebase domain

File Structure for Deployment

```
public/
├── index.html           (root file)
├── css/
│   ├── styles.css
│   └── animations.css
├── js/
│   ├── constants.js
│   ├── Pet.js
│   ├── Game.js
│   ├── StorageManager.js
│   ├── UIController.js
│   └── main.js
├── docs/                (optional - for documentation)
├── README.md
└── RESOURCES.md
```

Performance Optimization

Minimize Load Time

Gzip Compression:

- All text files compress 60-70%
- Server should gzip automatically

CSS Optimization:

- Total CSS: ~1.3MB (both files)

- Compresses to ~350KB gzipped
- Consider minification if needed

JavaScript Optimization:

- Total JS: ~1.8MB (all files)
- Compresses to ~450KB gzipped
- Runs efficiently in all browsers

HTML Optimization:

- Single HTML file: ~25KB
- Minimal DOM size
- Semantic HTML for fast parsing

Performance Metrics

- **Lighthouse Score:** Target 90+
 - **First Contentful Paint:** ~500ms
 - **Time to Interactive:** ~1s
 - **Memory Usage:** ~10MB
 - **Storage Used:** 1-5MB per player
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Security Features

Input Sanitization

All user inputs validated

No eval() or innerHTML for user data

XSS prevention through textContent

localStorage isolated to domain

HTTPS Requirements

- **Recommended:** Enable HTTPS on hosting
- **Security:** Encrypts data in transit
- **Certificates:** Free via Let's Encrypt
- **Browsers:** May warn without HTTPS

CORS Handling

- One cross-origin request: the Chat tab calls api.groq.com — requires a user-supplied Groq API key and only fires when the user sends a message
- All game data (stats, saves, expenses) is stored locally via localStorage — no data leaves the browser during normal gameplay
- Safe for same-origin deployment; the Groq endpoint handles its own CORS headers

Privacy

- No external tracking/analytics

- All data stored locally
- No user account required
- No data collection

Browser Support

Browser	Version	Support
Chrome	51+	Full
Firefox	54+	Full
Safari	10+	Full
Edge	15+	Full
Opera	38+	Full
IE 11	N/A	Not supported

Troubleshooting Deployment

Issue: Game won't load

Solution 1: Check file paths

- Verify css/js files are in correct directories
- Check browser console for 404 errors
- Ensure relative paths are correct

Solution 2: Check MIME types

- CSS files must have `text/css` MIME type
- JavaScript files must have `text/javascript` MIME type
- Most servers auto-detect correctly

Issue: Data not persisting

Solution 1: Check browser settings

- Private/Incognito mode disables localStorage
- Check browser storage limits
- Clear browser cache and reload

Solution 2: Check hosting permissions

- Some hosting restricts localStorage
- Try different hosting provider
- Use cookies as fallback (code modification)

Issue: Slow performance

Solution 1: Enable compression

- Server should gzip all responses
- Check hosting provider settings
- Use CDN for static assets

Solution 2: Check browser

- Close other tabs/extensions
- Clear browser cache
- Try different browser

Optional Enhancements

Add Custom Domain

1. **Buy Domain:** From registrar (GoDaddy, Namecheap, etc.)
2. **Configure DNS:** Point to hosting provider
3. **SSL Certificate:** Use Let's Encrypt (free)
4. **Update:** Domain links in deployment

Add Analytics (Optional)

```
<!-- Add to index.html if desired -->
<!-- Google Analytics or Plausible Analytics -->
<script async src="..."></script>
```

Add Service Worker (Progressive Web App)

```
// Create sw.js for offline support
if ('serviceWorker' in navigator) {
  navigator.serviceWorker.register('/sw.js');
}
```

Add Manifest (PWA Features)

```
{
  "name": "Virtual Pet Simulator",
  "short_name": "Pet Sim",
  "icons": [...],
  "start_url": "/index.html",
  "display": "standalone"
}
```

Long-Term Maintenance

Backup & Archival

- Keep source code in version control
- Backup regularly to cloud storage
- Archive with deployment documentation

Monitoring

- Check uptime if hosted
- Monitor error logs
- Track user feedback

Updates

- Security patches when needed
- Browser compatibility checks
- Feature additions as requested

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